

Classical General Relativity as a Non-Conservative Action-Dependent Field Theory

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Scaling symmetries have previously been examined for classical field theories described by singular Lagrangians; in this article, we apply these results to the first-order formulation of General Relativity. It is shown that the dynamical content of the Hilbert action may be formulated in terms of the conformal spacetime geometry, together with a dissipative sector, which is required in order to compensate the elimination of the notion of scale encoded by the conformal factor. Further, we consider the linearisation of the equations of motion of the scale-invariant action, demonstrating that the first-order metric perturbations satisfy a free wave equation, as expected. The second-order dynamics, describing gravitational backreaction, are found to be sourced by quadratic combinations of the first-order perturbations. However, these dynamics are non-conservative, as is made manifest by the presence of terms which couple the action sector with the geometrical degrees of freedom.

I. INTRODUCTION

As a classical field theory, General Relativity may be formulated in a variety of equivalent ways, each highlighting a different structural aspect of the theory. Amongst these diverse formulations, those in which the conformal mode - representing a local choice of ‘metre stick’ - is treated separately are particularly interesting, both in classical and quantum contexts [1, 2]. In such approaches, the remaining geometrical degrees of freedom are contained within a symmetric covariant tensor of rank two, whose components describe the conformal structure of the underlying spacetime.

In parallel, there has been growing interest in non-conservative extensions of classical field theory, such as those based on Herglotz variational principles [3], in which the action itself is incorporated into the space of dynamical variables. These frameworks are naturally able to capture dissipative or non-conservative effects, that cannot be replicated by adding terms to a standard Lagrangian. Such theories have recently been explored in gravitational settings [4, 5]; however, the nature of the action dependence of these models is relatively simplistic, and somewhat lacks physical motivation.

In this work, we combine the above-mentioned ideas, studying a reformulation of General Relativity, in which the conformal degree of freedom is identified as redundant, from the perspective of the closure of the algebra of dynamical observables. This observation allows us to eliminate the conformal factor, passing to a description which is dynamically equivalent to standard General Relativity, but exhibits a reduced gauge symmetry and a non-conservative structure. This is made manifest within both the Lagrangian and the corresponding field equations through their action dependence. Our objective is to present this novel construction, and to analyse its implications, which we do both generically at the level of the field theory, as well as in the particular setting of gravitational wave dynamics.

The first part of our work contains the required background material, with section (III) introducing the notion of scaling symmetries, and how the presence of such symmetries allows a reduction process to be carried out. Following this, sections (IV) and (V) detail the first-order formulation of General Relativity, expressed in frame field variables, together with the geometrical setting in which we shall make the symmetry reduction. The main results of our work, in which we obtain a first-order action-dependent description of General Relativity, are the subject of section (VI). As explained here, working with the first-order formalism is somewhat of a formal requirement, for the multisymplectic bundle corresponding to the velocity phase space is the *first* jet bundle over the space of fields. In this way, in order for the reduction process to be

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mathematically well-defined, we are in some sense obliged to work within the first-order formalism. However, having made the reduction, there is no impediment to our passing to the second-order metric formalism, which is both more familiar and convenient. Indeed, in section (VII), we recover the second-order formalism, expressed exclusively in terms of the metric and its derivatives. From the corresponding action, we compute the field equations, which are shown to contain precisely the same dynamical information as the more standard $G_{\mu\nu} = 0$ expression.

Finally, as an interesting application of our work, we consider the linearisation of the above-mentioned field equations around a flat background. In particular, we find that, at first order, the dynamics of the perturbation are unaffected by the action-dependent nature of our theory. The sole effect of this ‘dissipative’ sector is to reorganise the gauge degrees of freedom. At second order, by contrast, where the corresponding perturbation is associated with gravitational backreaction, we do encounter qualitative modifications from the presence of the action-density sector. Unlike the standard picture, in which the self-interaction of gravitational waves can be described in terms of an effective conserved energy-momentum tensor, here the dynamics are intrinsically non-conservative. The exchange of energy between the metric perturbations and the action degrees of freedom leads to a qualitatively different interpretation of gravitational wave propagation. We emphasise that such differences are wholly interpretational - our construction is, mathematically speaking, identical in dynamical content to standard General Relativity. We then conclude our work with a brief summary of the results obtained, together with a discussion of interesting lines of further investigation.

II. LEIBNIZ’S PRINCIPLE: THE IDENTITY OF INDISCERNIBLES

Whilst philosophical argumentation is frequently used in conjunction with physical insight, there are several notable instances in which the deduction of natural law has only been possible *because of* philosophical reasoning. In the current context, we refer to a principle due to Leibniz, known as the Identity of Indiscernibles [6]. This posits that if one has a theory that grants ontological distinction to two scenarios, for which the outcome of all measurements is identical, and whose differences lie solely within unobservable features, then this theory should be rejected, or at least considered non-fundamental. Such reasoning was employed most notably by Einstein in reference to accelerating frames of reference. In particular, the Equivalence Principle is an inevitable (albeit highly non-trivial) consequence of the refusal to grant ontological significance to unobservable differences between physical scenarios. An accelerating frame is indistinguishable from one which experiences a local gravitational field (of magnitude equal to the acceleration it induces), and so any theory which deems accelerating frames to be ontologically separate from those subject to a gravitational field should not be considered a faithful description of reality.

The recurrent success of this line of philosophical argumentation is the principal motivation behind our study of a class of symmetry known as *dynamical similarities* [7]. Mathematically, the deduction of the dynamics of a classical system may be considered an exercise in either functional extremisation or symplectic geometry; whilst not independent of one another, it is useful to consider each of them separately. From the extremisation perspective, it is often found that certain combinations of rescalings of the canonical variables (e.g $x \rightarrow \lambda^2 x$, $\dot{y} \rightarrow \lambda^{-1} \dot{y}$, for $\lambda \in \mathbb{R}$) have the sole effect of transforming the Lagrange density by an overall multiplicative factor ($\mathcal{L} \rightarrow \lambda^A \mathcal{L}$). Such a rescaled object clearly shares its extremal points with the original Lagrange density, and so describes the same classical dynamics. From the symplectic viewpoint, the rescaled variables are not merely a reparameterisation, or change of coordinates, but correspond to genuinely new trajectories on the system’s (velocity) phase space. These trajectories have the particular feature that they describe precisely the same observable dynamics as the original curves.

Following the principle of the Identity of Indiscernibles, a theory which possesses a scaling symmetry is not considered fundamental. As described in detail below, given a mathematical framework that is invariant under rescalings of (combinations of) its canonical variables, it is straightforward to construct a vector field which implements these transformations. In forming the quotient of the (velocity) phase space by the one-dimensional orbits of this vector field, we are able to eliminate this inherently non-Leibnizian feature of our theory, passing to a description in which each observable configuration evolves along its unique phase space trajectory. We refer to this formal procedure as a *contact reduction by scaling symmetry* [8]. We note that this quotient space is no longer symplectic, but inherits a contact structure. Indeed, the contact theories

that arise have the peculiar feature of being action-dependent, and are thus non-conservative in nature.

III. SYMMETRY REDUCTION OF MULTISYMPLECTIC FIELD THEORIES

In order to establish notational conventions used throughout, we begin with an account of the symmetry reduction of classical Lagrangian systems that admit dynamical similarities. A more detailed exposition of these ideas, together with their extension to the Hamiltonian formalism, may be found in [9]. Consider the fibre bundle $\pi : E \rightarrow M$, over the d -dimensional, orientable spacetime manifold M , with local coordinates x^μ ($\mu = 0, \dots, d-1$), and volume form $\omega = dx^0 \wedge \dots \wedge dx^{d-1} := d^d x$. The covariant configuration space E is of dimension $(d+n)$, and the first jet bundle $\kappa : J^1 E \rightarrow E$ of sections of π corresponds to the velocity phase space. We introduce the bundle projection $\hat{\pi} := \pi \circ \kappa : J^1 E \rightarrow M$, taking local coordinates on $J^1 E$ to be (x^μ, y^a, y_μ^a) , with $1 \leq a \leq n$. From this, it is clear that $J^1 E$ is of dimension $d+n(1+d)$. The Lagrangian density $\mathcal{L}$ is a $\hat{\pi}$ -semibasic d -form on $J^1 E$ [10], whose local coordinate expression is

$$\mathcal{L}(x^\mu, y^a, y_\mu^a) = L(x^\mu, y^a, y_\mu^a) \hat{\pi}^* \omega \quad (3.1)$$

in which $L : J^1 E \rightarrow \mathbb{R}$ refers to the Lagrangian function. The (pre-)multisymplectic form $\Omega_L \in \Omega^{d+1}(J^1 E)$ is defined to be $\Omega_L := -d\Theta_L$, with

$$\Theta_L = \frac{\partial L}{\partial y_\mu^a} dy^a \wedge d^{d-1} x_\mu - \left(\frac{\partial L}{\partial y_\mu^a} y_\mu^a - L \right) d^d x \quad (3.2)$$

For the purpose of implementing a contact reduction by scaling symmetry, it will be of benefit to introduce

$$\theta_L^\mu := -\iota_{\partial_{d-1}} \dots \iota_{\partial_0} (\Theta_L \wedge dx^\mu) = \frac{\partial L}{\partial y_\mu^a} dy^a \quad (3.3)$$

Given a (pre-)multisymplectic Lagrangian system $(J^1 E, \Omega_L)$, the equations of motion for holonomic, $\hat{\pi}$ -transverse multivector fields $\mathbf{X}_L \in \mathfrak{X}^d(J^1 E)$ are deduced from

$$\iota_{\mathbf{X}_L} \Omega_L = 0 \quad (3.4)$$

When our Lagrangian system is regular, Ω_L is a multisymplectic form, and multivector field solutions of (3.4) are guaranteed to exist, and are unique. Indeed, we have a somewhat stronger statement: the Lagrangian variational problem determines critical sections $\phi \in \Gamma(M, E)$, whose canonical lifting $j^1 \phi$ to $J^1 E$ are integral sections of a family of holonomic $\hat{\pi}$ -transverse multivector fields $\{\mathbf{X}_L\}$, each of which satisfies (3.4). Moreover, in a chart of local coordinates (x^μ, y^a, y_μ^a) on $J^1 E$, if $\phi(x) = (x^\mu, y^a(x))$, then $j^1 \phi(x) = \left(x^\mu, y^a(x), \frac{\partial y^a}{\partial x^\mu}(x) \right)$, with

$$\frac{\partial}{\partial x^\mu} \left(\frac{\partial L}{\partial y_\mu^a} \circ j^1 \phi \right) - \frac{\partial L}{\partial y^a} \circ j^1 \phi = 0 \quad (3.5)$$

When Ω_L is pre-multisymplectic, holonomic multivector field solutions are not guaranteed to exist; in the most favourable of cases, application of a constraint algorithm allows us to deduce the maximal submanifold $\mathcal{S}_f \hookrightarrow J^1 E$ upon which solutions do exist [11, 12]; integrability of these solutions is not guaranteed, and so must be examined as a final, step in the constraint analysis.

From the forms (3.3), we identify a vector field $\Sigma \in \mathfrak{X}^\infty(J^1 E)$ as a scaling symmetry of degree $\Lambda \in \mathbb{R}$ if

$$\mathfrak{L}_\Sigma L = \Lambda L \quad \mathfrak{L}_\Sigma \theta_L^\mu = \theta_L^\mu \quad \text{for } \mu = 0, \dots, d-1 \quad (3.6)$$

By means of a suitable change of variables, we adopt coordinates on $J^1 E$ adapted to the scaling direction, so that

$$\Sigma = \xi \frac{\partial}{\partial \xi} + \xi_\mu \frac{\partial}{\partial \xi_\mu} \quad (3.7)$$

In what follows, we shall focus on those cases in which ξ is a covariant configuration space variable. Such a requirement is not overly restrictive, and allows us to be far more general about the geometrical interpretation of the reduced system. Moreover, this is precisely the case we shall encounter with the Palatini action in section (VI). Defining $e^{\rho/\Lambda} = \xi$, we find that

$$L = e^{\rho} f(\rho_{\mu}, \phi^a, \phi_{\mu}^a) \quad \Sigma = \Lambda \frac{\partial}{\partial \rho} \quad (3.8)$$

in which ϕ^a are fields left unscaled by Σ , and ϕ_{μ}^a their corresponding velocities [9]. We now introduce

$$s^{\mu} := \frac{\partial f}{\partial \rho_{\mu}} \quad L^H := f - \rho_{\mu} s^{\mu} \quad (3.9)$$

The coordinate ρ is found to satisfy $\rho_{\mu} = -\frac{\partial L^H}{\partial s^{\mu}}$, and L^H defines an action-dependent multicontact Lagrangian. The variables s^{μ} are components of the action density, and confer frictional properties to the field theory described by L^H [13].

In general, multicontact (or Herglotz) Lagrangians have a similar geometrical description to that of the multisymplectic objects discussed so far. In particular, referring to the covariant velocity phase space $J^1 E$ from above, we define

$$\mathcal{S} := J^1 E \times_M \wedge^{d-1} T^* M \cong J^1 E \times \mathbb{R}^d \quad (3.10)$$

with projections $\tau : \mathcal{S} \rightarrow E$ and $\beta = \pi \circ \tau : \mathcal{S} \rightarrow M$. Local coordinates on $\mathcal{S}$ are $(x^{\mu}, y^a, y_{\mu}^a, s^{\mu})$, and a multicontact Lagrangian density is expressed as

$$\mathcal{L}(x^{\mu}, y^a, y_{\mu}^a, s^{\mu}) = L(x^{\mu}, y^a, y_{\mu}^a, s^{\mu}) \beta^* \omega \quad (3.11)$$

In analogy to (3.2), there exist forms $\Theta_L \in \Omega^d(\mathcal{S})$ and $\Omega_L \in \Omega^{d+1}(\mathcal{S})$, with

$$\Theta_L = \left(ds^{\mu} - \frac{\partial L}{\partial y_{\mu}^a} dy^a \right) \wedge d^{d-1} x_{\mu} + \left(\frac{\partial L}{\partial y_{\mu}^a} y_{\mu}^a - L \right) d^d x \quad (3.12)$$

$$\Omega_L = d\Theta_L - \frac{\partial L}{\partial s^{\mu}} dx^{\mu} \wedge \Theta_L \quad (3.13)$$

On $\mathcal{S}$, holonomic sections $\Psi : M \rightarrow \mathcal{S}$ is may be expressed locally as

$$\Psi(x) = \left(x^{\mu}, y^a(x), \frac{\partial y^a}{\partial x^{\mu}} \Big|_x, s^{\mu}(x) \right)$$

and the multicontact analogue of the Euler-Lagrange field equations (3.5) become

$$\begin{aligned} \frac{\partial}{\partial x^{\mu}} \left(\frac{\partial L}{\partial y_{\mu}^a} \circ \Psi \right) &= \left(\frac{\partial L}{\partial y^a} + \frac{\partial L}{\partial y_{\mu}^a} \frac{\partial L}{\partial s^{\mu}} \right) \circ \Psi \\ \frac{\partial s^{\mu}}{\partial x^{\mu}} &= L \circ \Psi \end{aligned} \quad (3.14)$$

The ideas presented on multicontact Lagrangians must be modified slightly for the case of L^H , obtained via the elimination of the scaling variable ρ . In particular, the change of variables $e^{\rho/\Lambda} = \xi$ required to render the multisymplectic Lagrangian of the form (3.8) clearly does not cover all of $J^1 E$, since $e^{\rho/\Lambda} > 0$.

Mathematically, ξ has been identified as a globally-defined object satisfying $\mathfrak{L}_{\Sigma} \xi = \xi$; such quantities are referred to as scaling functions. By writing $\xi = e^{\rho/\Lambda}$, we have revealed that the configuration space separates into two connected pieces $E_{\pm}$, with $E_{\pm} \cong \tilde{E} \times_M (M \times \mathbb{R}_{\pm})$. Here, $\tilde{E}$ is a codimension-1 subspace of E , interpreted as a configuration space composed of all field variables of E , except ξ . The trivial bundles $M \times \mathbb{R}_{\pm} \rightarrow M$ correspond to the scaling variable, taking either $\xi = +e^{\rho/\Lambda}$ or $\xi = -e^{\rho/\Lambda}$. Provided both components are accounted for, we may study a dynamically-equivalent theory, defined on the space

$$\mathcal{S} := J^1 \tilde{E} \times \mathbb{R}^d.$$

The Herglotz Lagrangian (3.9) is a function on $J^1 \tilde{E} \times \mathbb{R}^d$ and *not* on $J^1 \tilde{E}$ precisely because the scaling variable has been eliminated, whilst the d velocity coordinates ρ_μ have not. When $(J^1 E, \Theta_L)$ is a regular system, these coordinates - which, in the reduced theory, assume the role of the action density - generate a Reeb distribution, with the result that¹ $(\mathcal{S}, \Theta_{L^H})$ is a regular multicontact Lagrangian system, whose dynamics are deduced from (3.14).

The symmetry reduction of singular field theories has been studied extensively in [14]; it was found that when Θ_L is pre-multisymplectic, a number of additional criteria must be verified, else the form Θ_{L^H} calculated from the Herglotz Lagrangian does not endow the reduced space with a pre-multicontact structure. Supposing that $\mathcal{D} \subset \mathcal{TS}$ is some distribution, and that $\Xi := \beta^* \omega$ denotes the volume form on $\mathcal{S}$, we introduce the following set of q -forms

$$\text{Ann}^q(\mathcal{D}) := \{ \xi \in \Omega^q(\mathcal{S}) \mid \iota_X \xi = 0 \text{ for all } X \in \Gamma(\mathcal{D}) \} \quad (3.15)$$

The Reeb distribution associated with the triple $(\mathcal{S}, \Theta_L, \Xi)$ is then defined pointwise according to

$$\mathcal{D}_p^R := \{ X \in \ker \Xi_p \mid \iota_X d\Theta_{L^H} \in \text{Ann}_p^d(\ker \Xi) \} \quad (3.16)$$

Finally, with the characteristic distribution $\mathcal{C} := \ker \Xi \cap \ker \Theta_{L^H} \cap \ker d\Theta_{L^H}$, if Θ_{L^H} is to define a pre-multicontact structure on $\mathcal{S}$, we require that for some $0 < k \leq n(1+d)$, $\text{rank } \mathcal{D}^R = d+k$ and $\text{rank } \mathcal{C} = k$. The local Reeb vector fields R_μ are sections of $\mathcal{D}^R$, and satisfy $\iota_{R_\mu} \Theta_{L^H} = d^{d-1} x_\mu$.

The geometrical ideas developed over the course of this section will form the basis of our construction, and subsequent analysis of a contact-reduced first-order action for General Relativity. Before this, however, it will be of use to briefly review the description of gravitational actions using frame fields and spin connections, in both the first and second-order formalisms.

IV. ORTHONORMAL FRAMES AND SPIN CONNECTIONS

Given a d -dimensional manifold M , the frame bundle $Fr(TM) := FM \rightarrow M$ is a principal $GL(\mathbb{R}^d)$ -bundle, whose elements are pairs (x, Δ_x) , where Δ_x denotes a frame at the point $x \in M$. This frame is a set of d linearly independent sections $\Delta_x = (e_1|_x, \dots, e_d|_x)$, which provide a basis for the tangent space $T_x M$ [15]. It follows that, at each point, the tangent space admits two distinct bases: that provided by the elements e_I of the frame, and that of the coordinate derivatives ∂_μ . Here and throughout, $\mu, \nu, \dots$ are spacetime indices, while $I, J, \dots$ denote internal indices. There exist components $e_\mu^I(x)$ of a $GL(\mathbb{R}^d)$ matrix, such that $\partial_\mu = e_\mu^I(x) e_I$, and for some $v \in T_x M$, we may interchangeably write $v = v^\mu \partial_\mu$ or $v = v^I e_I$. An analogous statement holds for the cotangent bundle T^*M : at each point $x \in M$, we have a coordinate basis of 1-forms dx^μ , and a basis provided by the elements e^I of the coframe.

Let us consider the case of $d = 4$, and introduce a Lorentzian metric $g_{\mu\nu}$ of signature $(-, +, +, +)$ on M . When M is orientable, there is a reduction of the structure group of FM from $GL(\mathbb{R}^4)$ to $SO(1, 3)$, upon identifying the orientable orthonormal frames as a preferred subset. These satisfy $g(e_I, e_J) = \eta_{IJ}$, and we denote the subbundle of orientable orthonormal frames $SO(M) \subset FM$ [16, 17], constructing the associated bundle

$$E = SO(M) \times_{SO(1,3)} \mathbb{R}^{1,3} \quad (4.1)$$

in which we take the fundamental representation of $SO(1, 3)$. The orthonormal frame bundle $SO(M) \rightarrow M$ is a principal bundle, which admits a connection; this may be represented by a (unique) Lie algebra valued 1-form $\omega \in \Omega^1(SO(M), \mathfrak{so}(1, 3))$. Given some trivialising section $s : U \subset M \rightarrow SO(M)$, the quantity

¹ Here, Θ_{L^H} refers to the d -form on the reduced space, calculated according to (3.12).

$s^*\omega \in \Omega^1(U, \mathfrak{so}(1,3))$ is a local spin connection 1-form, which we denote ω_J^I . This connection naturally induces a corresponding structure on the associated bundle E , giving a covariant derivative $\mathcal{D}$, whose action on a vector $X = X^I e_I$ reads

$$\mathcal{D}X^I = dX^I + \omega_J^I X^J \quad (4.2)$$

Introducing local coordinates x^μ on M , we extend this definition to objects of arbitrary internal indices in the obvious manner; for example

$$\mathcal{D}_\mu X_J^I = \partial_\mu X_J^I + \omega_\mu^I{}_K X_J^K - \omega_\mu^K{}_J X_K^I \quad (4.3)$$

While $\mathcal{D}$ acts on internal indices, there is also a connection ∇ on the tangent bundle, which is sensitive only to spacetime indices

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\mu\lambda}^\nu V^\lambda \quad (4.4)$$

A priori, $\mathcal{D}$ and ∇ are completely independent, and neither is required to be metric-compatible. However, one obtains more physically interesting outcomes when additional conditions are imposed on these connections. Each has its own notion of torsion:

$$T_{\mu\nu}{}^\lambda := \Gamma_{\mu\nu}^\lambda - \Gamma_{\nu\mu}^\lambda \quad T^I := de^I + \omega_K^I \wedge e^K \quad (4.5)$$

which will often be taken to be vanishing. Finally, it is useful to introduce an extension of either $\mathcal{D}$ or ∇ to the space upon which it does not currently act, defining a ‘fully covariant derivative’. We adopt the following convention

$$\mathcal{D}_\mu V^{\lambda I} := \partial_\mu V^{\lambda I} + \Gamma_{\mu\rho}^\lambda V^{\rho I} + \omega_\mu^I{}_J V^{\lambda J} \quad (4.6)$$

V. GRAVITATIONAL ACTIONS

The conventional way in which one formulates a gravitational action is to employ the unique, torsion-free, metric-compatible connection, acting on sections of the tangent bundle. The resulting Lagrangian is a function of the metric tensor, together with its first and second derivatives [18]. This second-order formalism is not particularly amenable to a treatment based on multisymplectic geometry, as we would be required to work on the *second* jet bundle. For this reason, we shall favour the first-order formalism, and throughout, we follow the conventions of [19] with respect to index (anti)symmetrisation, taking $A^{(\mu\nu)} := A^{\mu\nu} + A^{\nu\mu}$ and $A^{[\mu\nu]} := A^{\mu\nu} - A^{\nu\mu}$.

Within the second-order framework, the connection on the tangent bundle is assumed to be metric-compatible ($\nabla_\alpha g_{\mu\nu} = 0$) and torsion-free ($T_{\mu\nu}{}^\lambda = 0$). The total covariant derivative $\mathcal{D}$ must also be compatible with the spacetime metric ($\mathcal{D}_\alpha g_{\mu\nu} = 0$), and with the internal Minkowski metric ($\mathcal{D}_\mu \eta_{IJ} = 0$). This latter condition implies that $\omega_{\mu IJ}$ (and by extension ω_μ^{IJ}) is antisymmetric in its Lie algebra indices. The above conditions are common to both the first and second-order formalisms; the distinguishing feature of a second-order action is the tetrad postulate

$$\mathcal{D}_\mu e_\nu^I = \partial_\mu e_\nu^I - \Gamma_{\mu\nu}^\lambda e_\lambda^I + \omega_\mu^I{}_J e_\nu^J \stackrel{!}{=} 0 \quad (5.1)$$

This condition identifies a unique, torsion-free spin connection, compatible with the frame field, which may be expressed as

$$\omega_\mu^{IJ} = \frac{1}{2} e^{\lambda[I} \left(\partial_{[\mu} e_{\lambda]}^{J]} + e^{\nu J]} e_\mu^K \partial_\nu e_{\lambda K} \right) \quad (5.2)$$

The curvature of the tangent bundle connection is an $\text{End}(TM)$ valued 2-form, which we denote r , so as to reserve R for that of the $SO(1,3)$ connection

$$\nabla_{[\mu} \nabla_{\nu]} V_\lambda = r_{\mu\nu\lambda}{}^\rho V_\rho + T_{\mu\nu}{}^\rho \nabla_\rho V_\lambda \quad (5.3)$$

For completeness, we have provided the generalisation for non-zero torsion. The curvature of the spin connection is an $\mathfrak{so}(1, 3)$ valued 2-form. Since $\mathcal{D}$ may act on both internal and spacetime indices, we have

$$\mathcal{D}_{[\mu}\mathcal{D}_{\nu]}V_\lambda = R_{\mu\nu\lambda}{}^\rho V_\rho + T_{\mu\nu}{}^\rho \mathcal{D}_\rho V_\lambda \quad \mathcal{D}_{[\mu}\mathcal{D}_{\nu]}V_I = R_{\mu\nu I}{}^J V_J + T_{\mu\nu}{}^\rho \mathcal{D}_\rho V_I \quad (5.4)$$

From a direct calculation with $T_{\mu\nu}{}^\rho = 0$, we see that

$$R_{\mu\nu}{}^{IJ} = \partial_{[\mu}\omega_{\nu]}{}^{IJ} + \omega_{[\mu}{}^{IK}\omega_{\nu]K}{}^J \quad (5.5)$$

and the Ricci scalar R reads

$$R = g^{\mu\nu} R_{\mu\nu} = e^\mu{}_I e^\nu{}_J R_{\mu\nu}{}^{IJ} \quad (5.6)$$

With this, the Einstein-Hilbert action is written as

$$S_{\text{EH}}[e] = \frac{1}{2\kappa} \int d^4x \, e \, e^\mu{}_I e^\nu{}_J R_{\mu\nu}{}^{IJ} \quad (5.7)$$

As a consequence of the tetrad postulate, S_{EH} takes the frame field as its sole dynamical variable. Relaxing the assumption that $\mathcal{D}_\mu e_\nu^I = 0$, we obtain a theory of two independent variables, in which the spin connection is no longer required to be torsion-free. This is precisely the content of the first-order formalism, for which the corresponding action - referred to as the *Hilbert-Palatini action* - is written as

$$S_{\text{HP}}[e, \omega] = \frac{1}{2\kappa} \int d^4x \, e \, e^\mu{}_I e^\nu{}_J R_{\mu\nu}{}^{IJ} \quad (5.8)$$

The variation of this action is a standard calculation [20, 21], and so we shall not belabour the point excessively; however, variation with respect to $\omega_\mu{}^{IJ}$ is illustrative, as it requires partial integration, which is relatively non-trivial for curved manifolds with torsion [22, 23]. Making use of

$$\delta e = -e \, e_\mu^I \delta e_\mu^I \quad (5.9)$$

variations with respect to the frame field read

$$\delta S_{\text{HP}} = \frac{1}{2\kappa} \int d^4x \, e \left(2e_K^\mu e_I^\lambda e^\nu{}_J R_{\lambda\nu}{}^{KJ} - e_I^\mu e_K^\lambda e^\rho{}_L R_{\lambda\rho}{}^{KL} \right) \delta e_\mu^I$$

and since $g_{\mu\nu} = e_\mu^I e_\nu^J \eta_{IJ}$, the bracketed term is easily manipulated into the familiar form

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0 \quad (5.10)$$

While identical in form to the vacuum field equations, e^I and ω^{IJ} are *independent*, and so the curvature R is that of the gauge connection, and *not* the tangent bundle. The second component of the calculation requires that we independently vary the spin connection; to this end, we note that, in general, with $\mathcal{D}_\alpha g_{\mu\nu} = 0$, we may make use of

$$\partial_\mu (e A^\mu B^\nu C_\nu) = e (\mathcal{D}_\mu A^\mu) B^\nu C_\nu + e A^\mu \mathcal{D}_\mu (B^\nu C_\nu) + e T_{\rho\mu}{}^\rho A^\mu B^\nu C_\nu \quad (5.11)$$

Setting the torsion to zero, and noting that variations of $\omega_\mu{}^{IJ}$ give

$$\delta R_{\mu\nu}{}^{IJ} = \partial_{[\mu} \delta \omega_{\nu]}{}^{IJ} + \delta \omega_{[\mu}{}^{IK} \omega_{\nu]K}{}^J + \omega_{[\mu}{}^{IK} \delta \omega_{\nu]K}{}^J = \mathcal{D}_{[\mu} \delta \omega_{\nu]}{}^{IJ} \quad (5.12)$$

we have

$$\delta S_{\text{HP}} = \frac{1}{2\kappa} \int d^4x \, e \, e^\mu{}_I e^\nu{}_J \mathcal{D}_{[\mu} \delta \omega_{\nu]}{}^{IJ} = -\frac{1}{2\kappa} \int d^4x \, e \, \mathcal{D}_\mu (e^\mu{}_I e^\nu{}_J) \delta \omega_\nu{}^{IJ} + \frac{1}{2\kappa} \int d^4x \, \partial_\mu (e e^\mu{}_I e^\nu{}_J \delta \omega_\nu{}^{IJ})$$

Discarding the total divergence, we have

$$\begin{aligned} R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} &= 0 \\ \mathcal{D}_\mu (e^\mu{}_I e^\nu{}_J) &= 0 \end{aligned} \quad (5.13)$$

Compatibility of the frame field is thus enforced dynamically, and implies that $R_{\mu\nu}$ and R coincide with the corresponding objects derived from the tangent bundle connection.

Thus far, we have presented the standard geometrical framework of the first-order action; in order to make contact with the ideas of section (III), we introduce the following covariant configuration space [24, 25]

$$\mathcal{E} := (T^*M \otimes E) \times_M C(SO(M)) \quad (5.14)$$

in which $C(SO(M)) \rightarrow M$ is the affine bundle of connections over M , modeled on the vector bundle $T^*M \times \text{Ad}(SO(M)) \rightarrow M$. Local coordinates on $J^1\mathcal{E}$ are then² $(x^\mu, e_\mu^I, \omega_\mu^{IJ}, \partial_\mu e_\nu^I, \partial_\mu \omega_\nu^{IJ})$, and we introduce the following projections

$$\Pi : E \rightarrow M \quad \pi : \mathcal{E} \rightarrow M \quad \kappa : J^1\mathcal{E} \rightarrow \mathcal{E} \quad \hat{\pi} = \pi \circ \kappa : J^1\mathcal{E} \rightarrow M$$

The Lagrangian density is a $\hat{\pi}$ -semibasic 4-form on $J^1\mathcal{E}$, given by

$$\mathcal{L}_{\text{HP}} = \frac{1}{2\kappa} e e_I^\mu e_J^\nu R_{\mu\nu}{}^{IJ} \hat{\pi}^* V \quad (5.15)$$

in which $V = dx^0 \wedge dx^1 \wedge dx^2 \wedge dx^3 := d^4x$ denotes the volume form on M .

VI. A FIRST-ORDER FRICTIONAL ACTION

In 3+1 spacetime dimensions, the metric possess ten independent components; nine of these describe the conformal geometry of the spacetime, while the tenth provides a notion of volumetric scale. It is possible to decompose the spacetime metric in a manner which reflects this distribution of the degrees of freedom, writing $g_{\mu\nu} = e^\phi h_{\mu\nu}$. The conformal factor $\phi(x)$ contains all information pertaining to scale, whilst $h_{\mu\nu}$ is a symmetric tensor of fixed determinant, often taken to be ± 1 , as befits the signature of g [26].

A. Constructing the Action-Dependent Theory

In what follows, we choose to decompose the tetrad e^I into a conformal factor, together with a field $\tilde{e}^I$, of fixed determinant: $\det \tilde{e}^I = -1$. In particular, we write

$$e_\mu^I = e^\phi \tilde{e}_\mu^I \quad e_I^\mu = e^{-\phi} \tilde{e}_I^\mu \quad (6.1)$$

Consider the effect of this decomposition on the torsion T^I

$$T^I = d(e^\phi \tilde{e}^I) + \omega_K^I \wedge (e^\phi \tilde{e}^K) = e^\phi \left[d\tilde{e}^I + \omega_K^I \wedge \tilde{e}^K + d\phi \wedge \tilde{e}^I \right] \quad (6.2)$$

On-shell, the torsion of ω_J^I is vanishing; comparison of this expression with (4.5) suggests that variations of the conformally-decomposed action will *not* identify ω_J^I as the connection compatible with the conformal frame field $\tilde{e}^I$. It would be natural, therefore, to seek a similarly scale-invariant object $\tilde{\omega}_J^I$, such that, in the same way that ω_J^I is dynamically fixed to be compatible with e^I , the modified connection $\tilde{\omega}_J^I$ should have an identical property, with respect to $\tilde{e}^I$. From above, we see that $\tilde{\omega}_J^I$ should satisfy

$$d\tilde{e}^I + \omega_K^I \wedge \tilde{e}^K = -d\phi \wedge \tilde{e}^I \quad \implies \quad d\tilde{e}^I + \tilde{\omega}_K^I \wedge \tilde{e}^K = 0 \quad (6.3)$$

A short calculation in local coordinates confirms that

$$\tilde{\omega}_\mu^{IJ} = \omega_\mu^{IJ} + \tilde{e}^{\lambda[I} \tilde{e}_\mu^{J]} \partial_\lambda \phi \quad (6.4)$$

² Strictly speaking, the notation employed here is incorrect. Local coordinates read $(x^\mu, e_\mu^I, \omega_\mu^{IJ}, e_{\mu,\nu}^I, \omega_{\mu,\nu}^{IJ})$. It is permissible to identify $e_{\mu,\nu}^I$ with $\partial_\nu e_\mu^I$ only when evaluated along the jet prolongation $j^1\phi$ of some $\phi \in \Gamma(M, \mathcal{E})$. For the sake of clarity, we persist with this slightly incorrect notation.

satisfies this requirement, and crucially, retains antisymmetry in its upper indices I and J . Expressing the action (5.8) in terms of the variables $(\phi, \tilde{e}^I, \tilde{\omega}_{IJ}^I)$, we find

$$S = \frac{1}{2\kappa} \int d^4x e^{2\phi} \left[\tilde{e}_I^\mu \tilde{e}_J^\nu \left(\tilde{R}_{\mu\nu}{}^{IJ} - \tilde{\mathcal{D}}_{[\mu} \left(\tilde{e}^{\lambda[I} \tilde{e}_{\nu]}^{J]} \partial_\lambda \phi \right) \right) - 6 \tilde{e}^{\mu K} \tilde{e}_K^\nu \partial_\mu \phi \partial_\nu \phi \right] \quad (6.5)$$

in which we have defined

$$\tilde{R}_{\mu\nu}{}^{IJ} := \partial_{[\mu} \tilde{\omega}_{\nu]}{}^{IJ} + \tilde{\omega}_{[\mu}{}^{IK} \tilde{\omega}_{\nu]K}{}^J$$

and $\tilde{\mathcal{D}}$ is a covariant derivative which acts only on $SO(1,3)$ indices, utilising the conformal spin connection; explicitly, for a tensor of mixed indices A_λ^{IJ}

$$\tilde{\mathcal{D}}_\mu A_\lambda^{IJ} := \partial_\mu A_\lambda^{IJ} + \tilde{\omega}_\mu{}^I{}_K A_\lambda^{KJ} + \tilde{\omega}_\mu{}^J{}_K A_\lambda^{IK}$$

At present, this action is not amenable to a contact reduction, due to the $\partial_\lambda \phi$ inside the covariant derivative; however, integrating by parts, discarding surface terms, and writing the conformal factor as $\Phi := 2\phi$, we find

$$S = \frac{1}{2\kappa} \int d^4x e^\Phi \left[\tilde{e}_I^\mu \tilde{e}_J^\nu \tilde{R}_{\mu\nu}{}^{IJ} + \tilde{e}^{\lambda[I} \tilde{e}_{\nu]}^{J]} \partial_\lambda \Phi \tilde{\mathcal{D}}_\mu (\tilde{e}_I^\mu \tilde{e}_J^\nu) + \frac{3}{2} \tilde{e}^{\mu K} \tilde{e}_K^\nu \partial_\mu \Phi \partial_\nu \Phi \right] \quad (6.6)$$

From the results of section (III), it should be clear that, at least structurally, the action is now of a form to which a contact reduction may be applied. Of course, it must first be verified that the conditions (3.6) are satisfied for the vector field $\Sigma = \partial_\Phi$. The forms θ_L^μ , defined in (3.3), are easily computed

$$\theta_L^\mu = \frac{e^\Phi}{2\kappa} \left[\left(\tilde{e}^{\mu[I} \tilde{e}_{\nu]}^{J]} \tilde{\mathcal{D}} (\tilde{e}_I^\mu \tilde{e}_J^\nu) + 3 \tilde{e}^{\mu K} \tilde{e}_K^\nu \partial_\nu \Phi \right) d\Phi + \partial_\lambda \Phi \left(\tilde{e}^{\lambda K} \tilde{e}_K^\mu \tilde{e}_\nu^I + 2 \tilde{e}^{\lambda I} \delta_\nu^\mu \right) d\tilde{e}_I^\nu + \tilde{e}_I^{[\mu} \tilde{e}_{\nu]}^{J]} d\omega_\nu{}^{IJ} \right]$$

from which it follows that $\Sigma = \partial_\Phi$ does indeed constitute a scaling symmetry of degree one. The Lagrangian

$$L(x^\mu, \tilde{e}^I, \tilde{\omega}_{IJ}^I, \Phi) = \frac{1}{2\kappa} e^\Phi \left(\tilde{e}_I^\mu \tilde{e}_J^\nu \tilde{R}_{\mu\nu}{}^{IJ} + \tilde{e}^{\lambda[I} \tilde{e}_{\nu]}^{J]} \partial_\lambda \Phi \tilde{\mathcal{D}}_\mu (\tilde{e}_I^\mu \tilde{e}_J^\nu) + \frac{3}{2} \tilde{e}^{\mu K} \tilde{e}_K^\nu \partial_\mu \Phi \partial_\nu \Phi \right) \quad (6.7)$$

is manifestly of the form $L = e^\Phi f$, and so we have

$$s^\mu = \frac{\partial f}{\partial (\partial_\mu \Phi)} = \frac{1}{2\kappa} \left[3 \tilde{e}^{\mu K} \tilde{e}_K^\nu \partial_\nu \Phi + \tilde{e}^{\mu[I} \tilde{e}_{\nu]}^{J]} \tilde{\mathcal{D}}_\rho (\tilde{e}_I^\rho \tilde{e}_J^\nu) \right] \quad (6.8)$$

The Herglotz Lagrangian is constructed as $L^H = f - (\partial_\mu \Phi) s^\mu$, and by inverting the above expression, we eliminate $\partial_\mu \Phi$ in favour of the action density. This calculation is somewhat lengthy, and offers little physical insight; we therefore omit the details, and affirm that the Herglotz Lagrangian has the following form

$$\begin{aligned} L^H = & \frac{1}{2\kappa} \tilde{e}_I^\mu \tilde{e}_J^\nu \tilde{R}_{\mu\nu}{}^{IJ} - \frac{\kappa}{3} \tilde{e}_\mu^K \tilde{e}_{\nu K} s^\mu s^\nu + \frac{1}{3} s^\mu \tilde{e}_\mu^{[I} \tilde{e}_\lambda^{J]} \tilde{\mathcal{D}}_\sigma (\tilde{e}_I^\sigma \tilde{e}_J^\lambda) \\ & - \frac{1}{12\kappa} \tilde{e}_\mu^{[I} \tilde{e}_\rho^{J]} \tilde{e}^{\mu[A} \tilde{e}_\lambda^{B]} \tilde{\mathcal{D}}_\sigma (\tilde{e}_I^\sigma \tilde{e}_J^\rho) \tilde{\mathcal{D}}_\alpha (\tilde{e}_A^\alpha \tilde{e}_B^\lambda) \end{aligned} \quad (6.9)$$

The first term of this Lagrangian is clearly a scale-free version of the original Palatini theory, while the second is quadratic on the action density. While the conformal factor *is* an empirically inaccessible degree of freedom, it does still possess an algebraic status within the Palatini action. Its excision does not, therefore, leave the rest of the Lagrangian unaffected. Indeed, as is ubiquitously the case with contact reduction, we eliminate a scaling degree of freedom at the cost of introducing friction, in the form of action-dependence. The remaining terms in (6.9) contain covariant derivatives of the frame fields, which vanish on-shell, as is shown in appendix A. From the results of section (III), we know that the Herglotz Lagrangian (6.9), together with the condition $\partial_\mu s^\mu = L^H$, faithfully reproduces the complete dynamical content of the original Palatini action, without reference to the scaling variable ϕ . We postpone further discussion of the equations of motion until section (VII).

While the Herglotz Lagrangian (6.9) is relatively unwieldy, the corresponding density may be cast exclusively in terms of forms, and is somewhat less complex

$$\mathcal{L}^H = \frac{1}{2\kappa} \star \Sigma^{IJ} \wedge \tilde{R}_{IJ} - \frac{\kappa}{3} (s \wedge \star s) + \frac{1}{3} (\star s) \wedge \star \iota_{\Sigma_{IJ}} \tilde{\mathcal{D}} \Sigma^{IJ} - \frac{1}{12\kappa} \iota_{\Sigma_{IJ}} \tilde{\mathcal{D}} \Sigma^{IJ} \wedge \star \iota_{\Sigma_{AB}} \tilde{\mathcal{D}} \Sigma^{AB} \quad (6.10)$$

Here, we have defined $\Sigma_{IJ} := \tilde{e}_I \wedge \tilde{e}_J$, and $\tilde{R}_{IJ} := \frac{1}{2} \tilde{R}_{\mu\nu IJ} dx^\mu \wedge dx^\nu$. The Hodge dual $\star$ should be understood to be taken with respect to the $\tilde{e}^I$.

B. Geometry of the Reduced Space

Having obtained the scale-invariant, frictional Lagrangian (6.9), we turn to a discussion of the geometry of the reduced space. Decomposing the frame field as $e^I = e^\phi \tilde{e}^I$ with a unit determinant condition on $\tilde{e}_\mu^I$ implies that the covariant configuration bundle $\mathcal{E}$ admits the following decomposition

$$\mathcal{E} \cong (M \times \mathbb{R}_+) \times_M (\tilde{E} \otimes T^*M) \times_M C(SO(M)) \quad (6.11)$$

Here, $\tilde{E}$ is a codimension-one subspace of E , such that sections of $\tilde{E} \otimes T^*M$ are the conformal frame fields $\tilde{e}^I$, and $M \times \mathbb{R}_+ \rightarrow M$ is a trivial bundle over M , corresponding to the scalar field ϕ . Expressed in this way, it is clear that the quotient space under the orbit of the scaling symmetry $\Sigma = \partial_\Phi$ is

$$\mathcal{E}_{\text{red}} \cong (\tilde{E} \otimes T^*M) \times_M C(SO(M))$$

As discussed extensively in [9] and [14], the elimination of the scaling variable does *not* remove the d independent velocity fields $\partial_\mu \Phi$, which assume the role of the action density s^μ . The Herglotz Lagrangian is thus a function on the space $J^1 \mathcal{E}_{\text{red}} \times \mathbb{R}^4$, with local coordinates

$$(x^\mu, \tilde{e}_\mu^I, \tilde{\omega}_\mu^{IJ}, \partial_\mu \tilde{e}_\nu^I, \partial_\mu \tilde{\omega}_\nu^{IJ}, s^\mu)$$

Referring to the discussion of section (III), when one carries out a contact reduction, the variable ξ typically assumes both positive and negative values. This requires particular care to be taken when writing³ $\xi = e^{\rho/\Lambda}$; if we do not also consider $\xi = -e^{\rho/\Lambda}$, some of the dynamical information of the original system is lost. Because our spacetime manifold is equipped with a Lorentzian metric of signature $(-, +, +, +)$, $\det g < 0$ at all points, and so the conformal decomposition $e^I = e^\phi \tilde{e}^I$ suffers no such $\pm$ ambiguities.

A further interesting feature of our scale-reduced theory concerns spacetime diffeomorphisms. In particular, under a change of local coordinates $x^\mu \rightarrow y^\mu(x)$ the frame field $\tilde{e}_\mu^I$ transforms as a covariant object. Any local coordinate transformation *must* preserve the condition $\det \tilde{e}^I = -1$, and so the spacetime symmetry group is reduced from $\text{Diff}(M)$ to the subgroup of unimodular diffeomorphisms, denoted $\text{Diff}_\mu(M)$. This unimodularity will be critical when considering the weak-field limit, and indeed more generally when analysing the significance of the structure of our theory.

VII. RECOVERING THE METRIC FORMALISM

Thus far, our analysis of General Relativity has been conducted exclusively within the first-order formalism, expressed in a non-coordinate basis. This has, in our opinion, been well-motivated: the first-order construction is required to work on the first jet bundle $J^1 \mathcal{E}$, while the use of the frame field and spin connection offers a more intuitive geometrical interpretation of the symmetry reduction, and is essential were we to add fermionic degrees of freedom. On the other hand, for practical applications, working with a non-coordinate basis is cumbersome; expressions quickly become unwieldy, and the algebra lengthy and tedious. It will therefore be of use to recast the results obtained into a more familiar form, employing the full spacetime

³ See discussion below (3.14) for notation.

metric. To this end, we write $g_{\mu\nu} = e^\Phi G_{\mu\nu}$, with $\det G_{\mu\nu} = -1$. On-shell, the connection compatible with g is that of Levi-Civita, whose corresponding connection coefficients are the Christoffel symbols. With the conformal decomposition of the metric, these now read

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2} G^{\alpha\rho} (\partial_\nu G_{\alpha\mu} + \partial_\mu G_{\nu\alpha} - \partial_\alpha G_{\mu\nu} + G_{\alpha\mu} \partial_\nu \Phi + G_{\nu\alpha} \partial_\mu \Phi - G_{\mu\nu} \partial_\alpha \Phi)$$

If we would like a connection $\tilde{\nabla}$, compatible with $G_{\mu\nu}$ in exactly the same way in which ∇ is compatible with $g_{\mu\nu}$, it would make sense to write

$$\tilde{\Gamma}_{\mu\nu}^\rho := \frac{1}{2} G^{\alpha\rho} (\partial_\nu G_{\alpha\mu} + \partial_\mu G_{\nu\alpha} - \partial_\alpha G_{\mu\nu}) \quad (7.1)$$

with the result that

$$\Gamma_{\mu\nu}^\rho = \tilde{\Gamma}_{\mu\nu}^\rho + \frac{1}{2} (\delta_\mu^\rho \partial_\nu \Phi + \delta_\nu^\rho \partial_\mu \Phi - G_{\mu\nu} \partial^\rho \Phi) \quad (7.2)$$

That this is a sensible construction can be seen by considering the metric-compatibility condition $\nabla_\alpha g_{\mu\nu} = 0$

$$\nabla_\alpha g_{\mu\nu} = \partial_\alpha g_{\mu\nu} - \Gamma_{\alpha\mu}^\rho g_{\rho\nu} - \Gamma_{\alpha\nu}^\rho g_{\mu\rho} = 0 \quad \xrightarrow{(7.2)} \quad \tilde{\nabla}_\alpha G_{\mu\nu} = \partial_\alpha G_{\mu\nu} - \tilde{\Gamma}_{\alpha\mu}^\rho G_{\rho\nu} - \tilde{\Gamma}_{\alpha\nu}^\rho G_{\mu\rho} = 0$$

This argument parallels the construction of the compatible spin connection $\tilde{\omega}^{IJ}$ in section (VI); indeed, by virtue of this analogy, it follows that the second-order Herglotz Lagrangian may be expressed as

$$\boxed{L^H = \frac{1}{2\kappa} \tilde{R}(G) - \frac{\kappa}{3} G_{\mu\nu} s^\mu s^\nu} \quad (7.3)$$

where the notation $\tilde{R}(G)$ is employed to emphasise that the curvature is a function of $G_{\mu\nu}$, together with its first and second derivatives. This expression, in conjunction with (6.10), constitutes the core result of our work. We have constructed an action for General Relativity which, as we proceed to show, despite making no reference to the conformal factor, correctly reproduces the complete dynamical content of the original theory.

Note that, for the connection $\tilde{\nabla}$, we have the following simplification

$$\tilde{\Gamma}_{\mu\nu}^\mu = \partial_\nu \ln \sqrt{-G} = 0$$

Of course, this also implies that $\tilde{\nabla}_\mu V^\mu = \partial_\mu V^\mu$ for any vector field V . Such observations lead to a number of simplifications; in particular, the coordinate expression for the Ricci tensor $\tilde{R}_{\mu\nu}$ is simply

$$\tilde{R}_{\mu\nu} = \partial_\rho \tilde{\Gamma}_{\mu\nu}^\rho - \tilde{\Gamma}_{\mu\lambda}^\rho \tilde{\Gamma}_{\rho\nu}^\lambda \quad (7.4)$$

The equations of motion of an action-dependent field theory may be obtained via variational methods [27]; the way in which such methods differ from those of conventional (action-independent) theories, is that the problem is now a constrained variational calculation. More precisely, the s^μ have the geometrical interpretation of components of a $(d-1)$ -form $s := s^\mu d^{d-1}x_\mu$; the field-theoretic generalisation of the Herglotz principle states that the integral of s over the boundary ∂D of some domain $D \subset M$ of the background spacetime must coincide with the value of the action of the solution. Symbolically, we have

$$\int_D \mathcal{L}^H \circ j^1 \phi = \int_{\partial D} s \quad (7.5)$$

in which $\mathcal{L}^H$ denotes the Herglotz Lagrange density, and the notation $\circ j^1 \phi$ refers to the evaluation of $\mathcal{L}^H$ along the 1st jet prolongation of the fields (c.f (3.5)). The Lagrange density, when evaluated on a given field configuration, is a top-form, and is thus closed. Locally, it follows that there exists a $(d-1)$ -form, whose differential coincides with the Lagrangian density: this is precisely s . Consequently, we have $\mathcal{L}^H \circ j^1 \phi = \partial_\mu s^\mu d^d x$; this constraint distinguishes non-conservative systems from their standard multisymplectic counterparts. Using a Lagrange multiplier, one typically introduces an extended action

$$\hat{S} = \int d^4x [(1-\lambda) \partial_\mu s^\mu + \lambda L^H] \quad (7.6)$$

which may be varied freely. From variations of the action density, one ascertains the equation governing the evolution of the Lagrange multiplier λ

$$\partial_\mu \lambda = -\lambda \frac{\partial L^H}{\partial s^\mu} \quad (7.7)$$

In the present case, this extended action fails to reproduce the correct field equations once a dynamical background metric is introduced. In particular, even though G is of fixed determinant, dynamical information is lost if this condition is not temporarily relaxed in the variational calculation. We thus reinstate the factor of $\sqrt{-G}$, and vary

$$\mathcal{L}^H = \sqrt{-G} \left[\frac{1}{2\kappa} \tilde{R}(G) - \frac{\kappa}{3} G_{\mu\nu} s^\mu s^\nu \right] d^4x \quad (7.8)$$

subject to the modified Herglotz constraint $\partial_\mu(\sqrt{-G} s^\mu) = \sqrt{-G} L^H$. There are two features of this calculation that we emphasise. Firstly, since we no longer require that $\sqrt{-G} = 1$, the curvature $\tilde{R}$ is not of the simplified form (7.4); since connection coefficients of the type $\tilde{\Gamma}_{\rho\nu}^\rho$ are no longer vanishing, we must use the full expression

$$\tilde{R}_{\mu\nu} = \partial_\rho \tilde{\Gamma}_{\mu\nu}^\rho - \partial_\mu \tilde{\Gamma}_{\rho\nu}^\rho + \tilde{\Gamma}_{\rho\lambda}^\rho \tilde{\Gamma}_{\mu\nu}^\lambda - \tilde{\Gamma}_{\mu\lambda}^\rho \tilde{\Gamma}_{\rho\nu}^\lambda$$

Additionally, despite having relaxed the condition $\sqrt{-G} = 1$, not all metric variations are admissible. We denote an allowed variation $\delta^A G_{\mu\nu}$, noting that such objects are characterised by the requirement that $\delta^A \sqrt{-G} = 0$. These restricted variations may be expressed in terms of an unconstrained $\delta G_{\mu\nu}$ according to

$$\delta^A G_{\mu\nu} = \delta G_{\mu\nu} - \frac{1}{4} G^{\alpha\beta} G_{\mu\nu} \delta G_{\alpha\beta} \quad (7.9)$$

It then follows that we may relate arbitrary and admissible variations of the action as

$$\frac{\delta S}{\delta^A G^{\mu\nu}} = \frac{\delta S}{\delta G^{\mu\nu}} - \frac{1}{4} G^{\mu\nu} G^{\alpha\beta} \frac{\delta S}{\delta G^{\alpha\beta}} \quad (7.10)$$

Since the metric requires that the Herglotz condition be modified to $\partial_\mu(\sqrt{-G} s^\mu) = \sqrt{-G} L^H$, we should consider an extended action of the form

$$\hat{S} := \int d^4x \sqrt{-G} \left[(1 - \lambda) \tilde{\nabla}_\mu s^\mu + \lambda L^H \right] \quad (7.11)$$

Here, there arises a small technical point, regarding the geometrical status of the action density variables, which does not affect our expression for the field equations, but should nevertheless be mentioned. As discussed above, the fields s^μ are the components of a $(d-1)$ -form, whose divergence $\partial_\mu s^\mu d^d x = \mathcal{L}^H \circ j^1 \phi$ is a metric-independent construction. However, once the Lagrangian density depends explicitly on a variational metric, and the action degrees of freedom are coupled to this metric (as is the case in (7.3)) the variational principle implicitly requires a volume form to define a diffeomorphism-invariant integral. Introducing the metric volume element $\sqrt{-G}$ alters the geometrical status of s^μ from a form density to the components of a vector field. Action-dependent field theories may equally well be formulated in a geometrical setting in which the s^μ correspond to components a vector field (see, for example [5]), and so the quantitative results obtained are unaltered.

Variation of the extended action (7.11) is a relatively straightforward calculation; making use of (7.10), we find that the frictional field equations read

$$\tilde{R}_{\mu\nu} - \frac{1}{4} \left(\tilde{R} + \frac{2\kappa^2}{9} G_{\alpha\beta} s^\alpha s^\beta - \frac{2\kappa}{3} \partial_\alpha s^\alpha \right) G_{\mu\nu} + \frac{2\kappa^2}{9} G_{\mu\alpha} G_{\nu\beta} s^\alpha s^\beta \quad (7.12)$$

$$- \frac{\kappa}{3} \left(G_{\mu\lambda} \tilde{\nabla}_\nu s^\lambda + G_{\lambda\nu} \tilde{\nabla}_\mu s^\lambda \right) = 0$$

$$\tilde{\nabla}_\mu s^\mu = \partial_\mu s^\mu = L^H \quad (7.13)$$

The steps leading to these frictional field equations resemble those used in the study of unimodular gravity (UG); indeed, the first term of (7.3) may be interpreted as a Lagrangian for just such a theory. However, it must be emphasised that our construction is *not* simply UG, and differs in several crucial regards. In the unimodular approach, one *chooses* to limit one's focus to metrics of unit determinant ab initio [28]. Our consideration of unimodular metrics, by contrast, was forced upon us after recognising that, of the ten independent degrees of freedom of $g_{\mu\nu}$, only those nine describing the conformal geometry of spacetime are truly indispensable from the perspective of the dynamical evolution of observable quantities. Despite being a redundant degree of freedom, the conformal factor does, of course, hold status within the mathematical structure of the original Hilbert action. As such, the effect of the removal of this degree of freedom is not simply to pass from the Hilbert action with g to the same action with G . Were this to be the case, our construction would indeed just yield UG. Instead, the removal of the conformal factor must be accompanied by a compensatory action dependence, whose presence is required in order to faithfully reproduce the original gravitational dynamics. This is precisely the content of the second term on the RHS of (7.3), and the feature which categorically separates our construction from UG.

Intuitively, at each point $x \in M$, the components $g_{\mu\nu}(x)$ may be considered to span a ten-dimensional space of symmetric matrices of Lorentzian signature $(-, +, +, +)$. In making the conformal decomposition, this factors locally into the product of an $\mathbb{R}_+$ and a nine-dimensional subspace, consisting of those Lorentzian metrics that have unit determinant. Quotienting by the orbits of the scaling symmetry eliminates precisely the $\mathbb{R}_+$, which we identify with a ‘radial’ direction within the ten-dimensional space. The remaining metric components are then specified exclusively in terms of ‘angles’. This is similar in spirit to how we would write $\mathbb{R}^{10} \setminus \{0\} \cong \mathbb{R}_+ \times S^9$; however, crucially, while S^9 is compact, the same is generally not true of the subspace $\{G_{\mu\nu} \mid \det G = -1\}$, and so this statement is merely an analogy.

As verification of the consistency of our formalism, one should carry out the variation of the full Hilbert action, expressed in conformal variables. Upon using the relation between the action density and field velocities $\partial_\mu \Phi$, which, in the metric formalism, reads

$$s^\mu = \frac{3}{2\kappa} G^{\mu\nu} \partial_\nu \Phi \quad (7.14)$$

the standard variational calculations which lead to the vacuum field equations should reproduce (7.12). This is indeed the case, and is demonstrated explicitly in appendix A.

As a final remark, we note that the construction presented thus far is valid in complete field-theoretic generality. Particular limiting cases have been studied in [13, 29] and [30], in the context of FLRW and (class-A) Bianchi cosmological models. It is a relatively straightforward exercise to make the appropriate simplifications for a spacetime that is homogeneous and/or isotropic. Taking $s^\mu = (S(t), 0, 0, 0)$ (as must be the case to recover the particle constraint $\dot{S} = L^H$) the field equations (7.12) are found to reduce to known results. Further, while we have focused exclusively on the vacuum case, our construction is perfectly amenable to the addition of (minimally-coupled) scalar fields, and so is fully compatible with, for example, models of inflationary cosmology.

VIII. THE WEAK-FIELD LIMIT

In the presence of weak gravitational fields, it is illustrative to study the linearised limit of General Relativity. Typically, one considers a spacetime metric $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, which differs from that of flat Minkowski space by some small perturbation $h_{\mu\nu}$. The field equations are then expanded to first order in $h_{\mu\nu}$, and the resulting theory found to describe the propagation of a symmetric rank-two covariant tensor field, on a flat background. More generally, when considering the linearised limit of a gravitational theory around any background spacetime, the perturbations are found to behave covariantly under the isometry group of that background. For the simple case of flat space, the isometry group corresponds to the Poincaré group in 3+1 dimensions. Having shown that General Relativity admits a scale-free description, it will be of interest to examine the linear limit of this theory. To this end, we introduce the following weak-field approximation

$$G_{\mu\nu} = \eta_{\mu\nu} + H_{\mu\nu} \quad G^{\mu\nu} = \eta^{\mu\nu} - H^{\mu\nu} \quad (8.1)$$

in which indices are raised and lowered with $\eta^{\mu\nu}$ and $\eta_{\mu\nu}$. Note that, in addition to being symmetric, our perturbation must also be traceless, since

$$\det(\eta_{\mu\nu} + H_{\mu\nu}) = -(1 + H) + \mathcal{O}(H^2) \quad H := \eta^{\mu\nu} H_{\mu\nu}$$

Action dependence of the Herglotz Lagrangian requires that we also expand the action density; for a flat background, the zeroth-order value of s^μ should be zero. Thus, to first order, we write $s^\mu = \sigma^\mu$. It is straightforward to expand (7.12) to first order in the perturbations; we find that

$$\frac{1}{2} (\partial_\rho \partial_\mu H^\rho_\nu + \partial_\rho \partial_\nu H^\rho_\mu - \square H_{\mu\nu}) - \frac{1}{2} \left(\partial_\alpha \partial_\beta H^{\alpha\beta} - \frac{2\kappa}{3} \partial_\alpha \sigma^\alpha \right) \eta_{\mu\nu} - \frac{\kappa}{3} (\eta_{\mu\lambda} \partial_\nu \sigma^\lambda + \eta_{\lambda\nu} \partial_\mu \sigma^\lambda) = 0 \quad (8.2)$$

Note that the first-order expansion of $\tilde{R}_{\mu\nu}$ does not contain the usual factor of $\partial_\mu \partial_\nu H$, as $H := \eta^{\mu\nu} H_{\mu\nu}$ is vanishing as a result of the unimodularity condition. At present, our theory is invariant under a large class of gauge transformations; in particular, infinitesimal diffeomorphisms generated by vector fields $\xi^\mu(x)$, such that $\partial_\mu \xi^\mu = 0$ do not alter the physical content of our description. Ordinarily, one uses part of this freedom to impose the harmonic gauge, which may be expressed as the requirement that $g^{\mu\nu} \Gamma_{\mu\nu}^\lambda = 0$. Rewriting this condition using the results of the previous section, we have

$$G^{\mu\nu} \tilde{\Gamma}_{\mu\nu}^\lambda - \frac{2\kappa}{3} s^\lambda = 0 \quad (8.3)$$

Expanding to first order, we lower the λ index with the Minkowski metric, obtaining

$$\partial_\rho H^\rho_\mu - \frac{2\kappa}{3} \eta_{\mu\lambda} \sigma^\lambda = 0 \quad (8.4)$$

A priori, it seems we have fixed more degrees of freedom than those afforded to us by unimodular diffeomorphisms. A vector field ξ^μ provides four independent choices; however, the divergence-free condition reduces this to just three. The expression (8.4) contains four components, and so appears excessively restrictive. The reconciliation lies in that not all four components of (8.4) are independent. Taking the derivative ∂^μ of the gauge-fixing condition, we obtain a term of the form $\partial_\mu \sigma^\mu$; this is subject to the Herglotz constraint, and so must coincide with L^H , expanded to first order. Consequently, (8.4) comprises only three independent degrees of freedom, and we may show explicitly how this gauge can be reached by considering the transformation of (8.4) under an infinitesimal shift $x^\mu \rightarrow x^\mu - \xi^\mu(x)$. On-shell, s^μ is a coordinate-dependent variable, and is thus subject to a transformation; to first order, however, this transformation is proportional to the zeroth-order value of s^μ , which has been set to zero. Consequently, we find that

$$\delta_\xi \left(\partial_\rho H^\rho_\mu - \frac{2\kappa}{3} \eta_{\mu\lambda} \sigma^\lambda \right) = \square \xi_\mu \quad \text{with } \partial_\mu \xi^\mu = 0$$

And thus our choice of gauge is accessible; with this, we see that (8.2) reduces to the standard free wave equation $\square H_{\mu\nu} = 0$. While this is not unexpected, there are a number of differences between the scale-reduced description and the conventional formulation of General Relativity. To appreciate these differences, we propose a plane wave solution of the form

$$H_{\mu\nu} = C_{\mu\nu} e^{ik \cdot x} \quad (8.5)$$

where $C_{\mu\nu}$ must be symmetric and traceless. Non-trivial solutions of the wave equation then require that $\eta_{\mu\nu} k^\mu k^\nu = 0$, as usual. Imposing the harmonic gauge condition (8.4), we have

$$ik_\rho C_\mu^\rho e^{ik \cdot x} = \frac{2\kappa}{3} \eta_{\mu\rho} \sigma^\rho \quad \implies \quad \sigma^\mu = \frac{3i}{2\kappa} C^{\mu\nu} k_\nu e^{ik \cdot x} \quad (8.6)$$

And so the frictional degrees of freedom must also behave in an oscillatory manner. When considering the standard formulation of General Relativity, one typically uses residual gauge symmetry to pass to the transverse-traceless gauge, so that $C_{\mu\nu} k^\nu = 0$ and $\eta^{\mu\nu} C_{\mu\nu} = 0$. Here, unimodular diffeomorphisms are insufficient to impose $C_{\mu\nu} k^\nu = 0$. However, we must also ensure that the Herglotz relation is satisfied order-by-order, and so we require that $\partial_\mu \sigma^\mu$ coincide with L^H expanded to first order

$$\partial_\mu \sigma^\mu \stackrel{!}{=} \frac{1}{2\kappa} \partial_\mu \partial_\nu H^{\mu\nu} \quad \implies \quad -\frac{3}{2\kappa} C^{\mu\nu} k_\mu k_\nu e^{ik \cdot x} = -\frac{1}{2\kappa} C^{\mu\nu} k_\mu k_\nu e^{ik \cdot x} \quad (8.7)$$

Consistency then requires that $C^{\mu\nu}k_\mu k_\nu = 0$. It would appear that, since $C^{\mu\nu}k_\mu k_\nu = 0$, but $C^{\mu\nu}k_\nu \neq 0$, we no longer obtain transverse modes. Such behaviour would be highly alarming, as despite eliminating the conformal degree of freedom, we have shown that the two theories are dynamically equivalent. The correct interpretation of our results is that the dissipative sector reorganises the gauge degrees of freedom. However, counting these degrees of freedom, we obtain the two propagating modes described in the standard framework. In particular, $C_{\mu\nu}$ is symmetric and traceless, and so possesses a total of nine independent components. A single unimodular diffeomorphism to enforce (8.4) eliminates three of these. The dynamically-determined consistency condition $C_{\mu\nu}k^\mu k^\nu = 0$ then brings the total down to five. Finally, there is residual gauge symmetry in the form of unimodular diffeomorphisms generated by vector fields ζ which satisfy $\partial_\mu \zeta^\mu = 0$ and $\square \zeta_\mu = 0$. Therefore the two propagating degrees of freedom we would expect from an analysis using standard General Relativity are reproduced here. In fact, we emphasise that this *must* be the case: while we have removed the conformal factor from our description, the underlying framework is, dynamically speaking, still General Relativity. As such, any apparent differences encountered because of the action-dependent nature of our theory are wholly interpretational; this description cannot (and must not) predict anything that is not contained within the standard framework.

Since the residual gauge symmetry satisfies the free wave equation, we write $\zeta^\mu = \epsilon^\mu e^{ik \cdot x}$, for some constant 4-vector ϵ^μ , with $k_\mu \epsilon^\mu = 0$. It then follows that

$$C_{\mu\nu} \mapsto \tilde{C}_{\mu\nu} = C_{\mu\nu} - i\epsilon_\mu k_\nu - i\epsilon_\nu k_\mu \quad (8.8)$$

Multiplying the above expression through by k^ν , we find that the quantity $C_{\mu\nu}k^\nu$ is invariant under residual gauge transformations, and so cannot simply be gauged away. This does *not* imply the presence of new degrees of freedom; instead, the behaviour of $C_{\mu\nu}k^\nu$ is governed dynamically, ensuring consistency with the expected number of propagating degrees of freedom. Thus, the elimination of the conformal factor changes nothing of the physical content of the theory, but does affect the way in which the constraints are distributed between gauge degrees of freedom and dynamics.

Let us now consider the field equations at second order; here, the frictional degrees of freedom cannot be decoupled from the wave dynamics by an astute choice of gauge, and our theory will be manifestly non-conservative. In order to be more concrete, we consider the following second-order terms

$$G_{\mu\nu} = \eta_{\mu\nu} + H_{\mu\nu} + \bar{H}_{\mu\nu} \quad s^\mu = \sigma^\mu + \bar{\sigma}^\mu \quad (8.9)$$

The second-order expansion of (7.12) is expressed schematically as

$$\Gamma_{\mu\nu}^{(1)}[\eta + H, \sigma] + \Gamma_{\mu\nu}^{(2)}[\eta + \bar{H}, \bar{\sigma}] = 0 \quad (8.10)$$

in which $\Gamma_{\mu\nu}^{(1)}[\eta + \bar{H}, \bar{\sigma}]$ refers to those terms linear in the second-order parameters, and $\Gamma_{\mu\nu}^{(2)}[\eta + H, \sigma]$ contains terms quadratic in $H_{\mu\nu}$ and σ^μ . After some work, we find that $\Gamma_{\mu\nu}^{(1)}[\eta + \bar{H}, \bar{\sigma}]$ is given by

$$\begin{aligned} \Gamma_{\mu\nu}^{(1)}[\eta + \bar{H}, \bar{\sigma}] = & \frac{1}{2} (\partial_\rho \partial_\mu \bar{H}^\rho_\nu + \partial_\rho \partial_\nu \bar{H}^\rho_\mu - \partial_\mu \partial_\nu \bar{H} - \square \bar{H}_{\mu\nu}) - \frac{1}{4} \left(\partial_\alpha \partial_\beta \bar{H}^{\alpha\beta} - \square \bar{H} - \frac{2\kappa}{3} \partial_\alpha \bar{\sigma}^\alpha \right) \eta_{\mu\nu} \\ & - \frac{\kappa}{3} (\eta_{\mu\lambda} \partial_\nu \bar{\sigma}^\lambda + \eta_{\lambda\nu} \partial_\mu \bar{\sigma}^\lambda) \end{aligned} \quad (8.11)$$

Note that, unlike the case of the first-order expansion, here $\bar{H} \neq 0$. The second-order piece $\Gamma_{\mu\nu}^{(2)}[\eta + H, \sigma]$ contributes the following terms

$$\begin{aligned} \Gamma_{\mu\nu}^{(2)}[\eta + H, \sigma] = & \tilde{R}_{\mu\nu}^{(2)} - \frac{1}{4} \left(\tilde{R}^{(2)} - \frac{1}{2} H^{\alpha\beta} \left(2\partial_\rho \partial_\alpha H^\rho_\beta - \square H_{\alpha\beta} \right) + \frac{2\kappa^2}{9} \eta_{\alpha\beta} \sigma^\alpha \sigma^\beta \right) \eta_{\mu\nu} \\ & - \frac{\kappa}{3} (\sigma^\rho \partial_\rho H_{\mu\nu} + H_{\mu\rho} \partial_\nu \sigma^\rho + H_{\rho\nu} \partial_\mu \sigma^\rho) + \frac{2\kappa^2}{9} \eta_{\mu\alpha} \eta_{\nu\beta} \sigma^\alpha \sigma^\beta \end{aligned} \quad (8.12)$$

in which $\tilde{R}_{\mu\nu}^{(2)}$ denotes those terms in the expansion of the Ricci tensor which contain terms schematically of the form $\partial H \partial H$ and $H \partial \partial H$, and $\tilde{R}^{(2)} := \eta^{\mu\nu} \tilde{R}_{\mu\nu}^{(2)}$. In the above expansion, we have also omitted a term

proportional to the gauge condition (8.4). The full field equations expanded to second order are simply the sum of (8.11) and (8.12); the second-order gauge condition may be conveniently expressed as

$$\partial_\rho \bar{H}_\mu^\rho - \frac{2\kappa}{3} \eta_{\mu\rho} \bar{\sigma}^\rho = \frac{1}{2} \partial_\mu \bar{H} + H^{\alpha\beta} \partial_\beta H_{\mu\alpha} - \frac{1}{2} H^{\alpha\beta} \partial_\mu H_{\alpha\beta} + H_\mu^\alpha \partial_\beta H_\alpha^\beta \quad (8.13)$$

With this, the final second-order expansion becomes

$$\begin{aligned} \square \bar{H}_{\mu\nu} - \frac{1}{4} \square \bar{H} \eta_{\mu\nu} + \frac{2\kappa}{3} (\sigma^\lambda \partial_\lambda H_{\mu\nu} + H_{\mu\lambda} \partial_\nu \sigma^\lambda + H_{\lambda\nu} \partial_\mu \sigma^\lambda) &= Q_{\mu\nu} + C \eta_{\mu\nu} \\ &+ \frac{4\kappa^2}{9} \left(\eta_{\mu\alpha} \eta_{\beta\nu} - \frac{1}{4} \eta_{\mu\nu} \eta_{\alpha\beta} \right) \sigma^\alpha \sigma^\beta \end{aligned} \quad (8.14)$$

Here, we have collected all terms of the form $\partial H \partial H$ and $H \partial \partial H$ into the object $Q_{\mu\nu}$, which includes the second-order expansion $\tilde{R}_{\mu\nu}^{(2)}$. Similarly, terms quadratic in H which have no free indices have been grouped into the object we have called C .

Physically, the second-order expansion describes the self-interaction of the propagating modes. The first two terms on the left-hand side of (8.14) correspond to the standard kinetic structure, modified by the unimodularity condition. The remaining terms, involving σ^μ , reflect a non-trivial coupling between the wave degrees of freedom and the action-density sector, signaling the intrinsically non-conservative nature of the theory. In contrast with the standard framework of General Relativity, where the second-order dynamics may be interpreted in terms of an effective conserved energy-momentum tensor for gravitational waves, here the backreaction is not governed by a conserved quantity. Instead, energy is exchanged between the geometric perturbations and the additional degrees of freedom encoded in s^μ . This provides a dynamical realisation of the fact that, following the elimination of the conformal mode, the constraints of the theory are redistributed between gauge symmetry and dynamics.

IX. DISCUSSION AND CONCLUSIONS

In this work, we have analysed a reformulation of classical General Relativity, in which the conformal degree of freedom has been identified as redundant. The evolution of the remaining dynamical variables may be faithfully reproduced without reference to this ontologically-inaccessible parameter. While dynamically equivalent to standard General Relativity, our contact-reduced theory exhibits a reduced gauge symmetry, being invariant only under volume-preserving diffeomorphisms, and requires a qualitative reinterpretation, as an intrinsically non-conservative description, which is made manifest by an action-dependent Lagrangian.

While the first-order formalism provided the correct geometrical setting in which to make the symmetry reduction, the slightly unwieldy nature of the frame field algebra led us to recast our results in terms of a second-order metric theory. Such a change in description was particularly appropriate for our subsequent study of the weak-field limit, in which we linearised the field equations around a flat Minkowski background. It was shown that, at the level of linear perturbations, the theory propagates the expected two physical degrees of freedom. However, the reduced gauge symmetry prevents the imposition of full transversality, and the elimination of non-physical components is achieved through a combination of residual gauge freedom and dynamical constraints.

At second order, the differences became more pronounced; we observed that the backreaction of gravitational waves was no longer governed by an effective conserved energy-momentum tensor, but instead involved a non-trivial coupling to the action-density sector. This led to a departure from the standard interpretation of gravitational wave energy, and reflects the fundamentally non-conservative nature of the theory.

One particularly interesting aspect of our construction lies in the fact that we are able to reproduce the complete (observable) dynamical content of General Relativity, as captured by the Hilbert action, without need to reference the conformal factor ϕ . Indeed, this degree of freedom is frequently assigned the physical interpretation of a volume/scale parameter. In such cases, this interpretation becomes highly non-physical when we approach the initial singularity, where the standard mathematical framework becomes pathological. It was shown in [30] that for the simple case of a matter-sourced FLRW model, while the conventional

description presents non-predictive behaviour at the initial singularity, the contact-reduced model suffered no such pathologies. Indeed, it was found that the solution could be continued in a perfectly predictive manner, and that traversing the singularity was associated with a change in the orientation of the spacetime manifold. The results of the present work constitute a complete field-theoretic generalisation of the contact reduction of classical General Relativity, and so naturally subsume the content of [30]. It would be of great interest, therefore, to examine whether techniques similar to those used for the particular case of homogeneous cosmologies may be utilised to draw more general conclusions about the nature of General Relativity in the vicinity of singularities, in which spacetime volumes collapse to zero.

In [14], it was found that the processes of contact reduction and phase space restriction are commutative. This is not surprising; however, it raises the interesting possibility of performing a $3 + 1$ decomposition of the contact-reduced Lagrangian (7.3), passing to a contact Hamiltonian description. It would be illustrative to examine whether any components of the constraint analysis are simplified by working directly with the reduced ontology, which makes no reference to the conformal factor. Indeed, more broadly, our work highlights how dynamically equivalent formulations of a single classical theory can offer novel insight, and in some cases/regimes - as exemplified by our discussion of singularities - be a more suitable description than the standard framework.

Appendix A: Frictional Field Equations

Upon excising the conformal factor from our ontology, we obtained a first-order action-dependent Lagrangian of the form

$$L^H = \frac{1}{2\kappa} \tilde{e}_I^\mu \tilde{e}_J^\nu \tilde{R}_{\mu\nu}{}^{IJ} - \frac{\kappa}{3} \tilde{e}_\mu^K \tilde{e}_{\nu K} s^\mu s^\nu + \frac{1}{3} s^\mu \tilde{e}_\mu^{[I} \tilde{e}_\lambda^{J]} \tilde{\mathcal{D}}_\sigma (\tilde{e}_I^\sigma \tilde{e}_J^\lambda) - \frac{1}{12\kappa} \tilde{e}_\mu^{[I} \tilde{e}_\rho^{J]} \tilde{e}^{\mu[A} \tilde{e}_\lambda^{B]} \tilde{\mathcal{D}}_\sigma (\tilde{e}_I^\sigma \tilde{e}_J^\rho) \tilde{\mathcal{D}}_\alpha (\tilde{e}_A^\alpha \tilde{e}_B^\lambda) \quad (\text{A1})$$

We shall demonstrate that, as claimed in the discussion surrounding (6.9), the terms containing covariant derivatives of the frame fields vanish on-shell. Having done so, we shall pass to the second-order metric formalism, proving that variation of the multisymplectic action (5.7), expressed in conformal variables, does indeed reproduce the field equations (7.12).

As discussed in the main text, action-dependent theories may be treated as unconstrained variational problems, provided a Lagrange multiplier is inserted to enforce the Herglotz condition. Since we consider only variations of $\tilde{\omega}^{IJ}$, and not of the frame field, we need not concern ourselves with any of the subtleties related to the unit-determinant condition (see discussion surrounding (7.8)). It therefore suffices to consider

$$\hat{S} := \int d^4x \left[(1 - \lambda) \partial_\mu s^\mu + \lambda L^H \right]$$

An expression for the derivative $\partial_\mu \lambda$ of the Lagrange multiplier will be required, whenever partial integration is carried out; varying $\hat{S}$ with respect to s^μ , we find that

$$\begin{aligned} \delta \hat{S} &= \int d^4x \left[(1 - \lambda) \delta(\partial_\mu s^\mu) + \frac{\lambda}{3} \left(-2\kappa \tilde{e}_\mu^K \tilde{e}_{\nu K} s^\nu + \tilde{e}_\mu^{[I} \tilde{e}_\lambda^{J]} \tilde{\mathcal{D}}_\sigma (\tilde{e}_I^\sigma \tilde{e}_J^\lambda) \right) \delta s^\mu \right] \\ &= \int d^4x \left[\partial_\mu \lambda - \frac{\lambda}{3} \left(2\kappa \tilde{e}_\mu^K \tilde{e}_{\nu K} s^\nu - \tilde{e}_\mu^{[I} \tilde{e}_\lambda^{J]} \tilde{\mathcal{D}}_\sigma (\tilde{e}_I^\sigma \tilde{e}_J^\lambda) \right) \right] \delta s^\mu \end{aligned}$$

from which it follows that

$$\partial_\mu \lambda = \frac{\lambda}{3} \left(2\kappa \tilde{e}_\mu^K \tilde{e}_{\nu K} s^\nu - \tilde{e}_\mu^{[I} \tilde{e}_\lambda^{J]} \tilde{\mathcal{D}}_\sigma (\tilde{e}_I^\sigma \tilde{e}_J^\lambda) \right) \quad (\text{A2})$$

We may now consider varying $\hat{S}$ with respect to $\tilde{\omega}^{IJ}$. The second term on the right-hand side of (A1) contains no reference to the spin connection; it will be of benefit to expand each of the covariant derivative

terms, making explicit their dependence on $\tilde{\omega}^{IJ}$

$$\begin{aligned} L^H \supset & \frac{1}{2\kappa} \tilde{e}_I^\mu \tilde{e}_J^\nu \tilde{R}_{\mu\nu}{}^{IJ} - \frac{1}{3} s^\mu \tilde{e}_{\mu[I} \tilde{e}_{\rho J]} (\tilde{\omega}_\sigma{}^{KI} \tilde{e}_K^\sigma \tilde{e}^{\rho J} + \tilde{\omega}_\sigma{}^{KJ} \tilde{e}^{\sigma I} \tilde{e}_K^\rho) \\ & + \frac{1}{12\kappa} \tilde{e}_{\mu[I} \tilde{e}_{\rho J]} \tilde{e}^{\mu[A} \tilde{e}_\lambda^{B]} (\tilde{\omega}_\sigma{}^{KI} \tilde{e}_K^\sigma \tilde{e}^{\rho J} + \tilde{\omega}_\sigma{}^{KJ} \tilde{e}^{\sigma I} \tilde{e}_K^\rho) \tilde{\mathcal{D}}_\alpha (\tilde{e}_A^\alpha \tilde{e}_B^\lambda) \end{aligned} \quad (A3)$$

Note that, in expanding the covariant derivatives, we have intentionally omitted any terms such as $\partial_\sigma (\tilde{e}_I^\sigma \tilde{e}_J^\lambda)$, as these do not contribute to our calculation. We have also swapped upper and lower I and J indices, as compared to (A1). Not forgetting the overall multiplicative factor of λ , variations of $\hat{S}$ with respect to $\tilde{\omega}_\mu{}^{IJ}$ yield

$$\begin{aligned} \delta \hat{S} &= \int d^4x \lambda \left[\frac{1}{2\kappa} \tilde{e}_I^{[\mu} \tilde{e}_J^{\nu]} \tilde{\mathcal{D}}_\mu \delta \tilde{\omega}_\nu{}^{IJ} - \frac{1}{3} s^\mu \tilde{e}_{\mu[I} \tilde{e}_{\rho J]} (\tilde{e}_K^\sigma \tilde{e}^{\rho J} \delta \tilde{\omega}_\sigma{}^{KI} + \tilde{e}^{\sigma I} \tilde{e}_K^\rho \delta \tilde{\omega}_\sigma{}^{KJ}) \right. \\ &\quad \left. + \frac{1}{6\kappa} \tilde{e}_{\mu[I} \tilde{e}_{\rho J]} \tilde{e}^{\mu[A} \tilde{e}_\lambda^{B]} (\tilde{e}_K^\sigma \tilde{e}^{\rho J} \delta \tilde{\omega}_\sigma{}^{KI} + \tilde{e}^{\sigma I} \tilde{e}_K^\rho \delta \tilde{\omega}_\sigma{}^{KJ}) \tilde{\mathcal{D}}_\alpha (\tilde{e}_A^\alpha \tilde{e}_B^\lambda) \right] \\ &= \int d^4x \lambda \left[\frac{1}{2\kappa} \tilde{e}_I^{[\mu} \tilde{e}_J^{\nu]} \tilde{\mathcal{D}}_\mu \delta \tilde{\omega}_\nu{}^{IJ} - \frac{1}{3} \tilde{e}_{\mu[I} \tilde{e}_{\rho J]} \left(s^\mu - \frac{1}{2\kappa} \tilde{e}^{\mu[A} \tilde{e}_\lambda^{B]} \tilde{\mathcal{D}}_\alpha (\tilde{e}_A^\alpha \tilde{e}_B^\lambda) \right) (\tilde{e}_K^\sigma \tilde{e}^{\rho J} \delta \tilde{\omega}_\sigma{}^{KI} + \tilde{e}^{\sigma I} \tilde{e}_K^\rho \delta \tilde{\omega}_\sigma{}^{KJ}) \right] \end{aligned}$$

We recognise the bracketed term containing s^μ as a rearrangement of (6.8). In order to simplify the action further, we observe that

$$\tilde{e}_{\mu[I} \tilde{e}_{\rho J]} (\tilde{e}_K^\sigma \tilde{e}^{\rho J} \delta \tilde{\omega}_\sigma{}^{KI} + \tilde{e}^{\sigma I} \tilde{e}_K^\rho \delta \tilde{\omega}_\sigma{}^{KJ}) = 2 \tilde{e}_{\mu I} \tilde{e}_K^\sigma \delta \tilde{\omega}_\sigma{}^{KI}$$

Inserting this into our expression for $\delta \hat{S}$, we have

$$\delta \hat{S} = \int d^4x \lambda \left[\frac{1}{2\kappa} \tilde{e}_I^{[\mu} \tilde{e}_J^{\nu]} \tilde{\mathcal{D}}_\mu \delta \tilde{\omega}_\nu{}^{IJ} - \frac{2}{3} \tilde{e}_{\mu I} \tilde{e}_K^\sigma \left(s^\mu - \frac{1}{2\kappa} \tilde{e}^{\mu[A} \tilde{e}_\lambda^{B]} \tilde{\mathcal{D}}_\alpha (\tilde{e}_A^\alpha \tilde{e}_B^\lambda) \right) \delta \tilde{\omega}_\sigma{}^{KI} \right]$$

This expression is now of a form that may be integrated by parts; in particular, we integrate the first term, making use of (A2) to rewrite $\partial_\mu \lambda$ as something proportional to λ

$$\begin{aligned} \delta \hat{S} &= - \int d^4x \lambda \left[\frac{1}{2\kappa} \left(\frac{1}{3} \tilde{e}_I^{[\mu} \tilde{e}_J^{\nu]} \left(2\kappa \tilde{e}_\mu{}^K \tilde{e}_{\lambda K} s^\lambda - \tilde{e}_\mu^{[A} \tilde{e}_\lambda^{B]} \tilde{\mathcal{D}}_\sigma (\tilde{e}_A^\sigma \tilde{e}_B^\lambda) \right) + \tilde{\mathcal{D}}_\mu (\tilde{e}_I^{[\mu} \tilde{e}_J^{\nu]}) \right) \delta \tilde{\omega}_\nu{}^{IJ} \right. \\ &\quad \left. + \frac{2}{3} \tilde{e}_{\mu I} \tilde{e}_K^\sigma \left(s^\mu - \frac{1}{2\kappa} \tilde{e}^{\mu[A} \tilde{e}_\lambda^{B]} \tilde{\mathcal{D}}_\alpha (\tilde{e}_A^\alpha \tilde{e}_B^\lambda) \right) \delta \tilde{\omega}_\sigma{}^{KI} \right] \\ &= - \int d^4x \lambda \left[\frac{2}{3} \tilde{e}_{\mu I} \tilde{e}_K^\nu \left(s^\mu - \frac{1}{2\kappa} \tilde{e}^{\mu[A} \tilde{e}_\lambda^{B]} \tilde{\mathcal{D}}_\sigma (\tilde{e}_A^\sigma \tilde{e}_B^\lambda) \right) \delta \tilde{\omega}_\nu{}^{KI} + \frac{1}{2\kappa} \tilde{\mathcal{D}}_\mu (\tilde{e}_I^{[\mu} \tilde{e}_J^{\nu]}) \delta \tilde{\omega}_\nu{}^{IJ} \right. \\ &\quad \left. + \frac{2}{3} \tilde{e}_{\mu I} \tilde{e}_K^\sigma \left(s^\mu - \frac{1}{2\kappa} \tilde{e}^{\mu[A} \tilde{e}_\lambda^{B]} \tilde{\mathcal{D}}_\alpha (\tilde{e}_A^\alpha \tilde{e}_B^\lambda) \right) \delta \tilde{\omega}_\sigma{}^{KI} \right] \end{aligned}$$

It is then clear that, upon swapping the upper I and K indices, the last term cancels with the first, leaving

$$\delta \hat{S} = - \int d^4x \frac{1}{2\kappa} \lambda \tilde{\mathcal{D}}_\mu (\tilde{e}_I^{[\mu} \tilde{e}_J^{\nu]}) \delta \tilde{\omega}_\nu{}^{IJ} = 0$$

from which it immediately follows that

$$\tilde{\mathcal{D}}_\mu (\tilde{e}_I^{[\mu} \tilde{e}_J^{\nu]}) = 0 \quad (A4)$$

With this, as claimed, the second-order Herglotz Lagrangian (A1) reduces to

$$L^H = \frac{1}{2\kappa} \tilde{e}_I^\mu \tilde{e}_J^\nu \tilde{R}_{\mu\nu}{}^{IJ} - \frac{\kappa}{3} \tilde{e}_\mu{}^K \tilde{e}_{\nu K} s^\mu s^\nu \quad (A5)$$

In section (VII), we shifted the focus of our analysis, favouring the metric formalism. The second-order Lagrangian (A5), expressed in terms of $G_{\mu\nu}$ and s^μ , reads

$$L^H = \frac{1}{2\kappa} \tilde{R} - \frac{\kappa}{3} G_{\mu\nu} s^\mu s^\nu \quad (A6)$$

As demonstration of the consistency of our formalism, we would like to show that the full Hilbert action, when expressed in conformal variables, leads precisely to the field equations (7.12). In doing so, we will have shown explicitly that no dynamical information is lost when eliminating the scaling degree of freedom, and that, as claimed, the conformal factor is redundant from the perspective of observable dynamics. We begin by noting that, in d dimensions, two metrics g and G related by a conformal rescaling $g_{\mu\nu} = e^\Phi G_{\mu\nu}$ have Ricci scalars R and $\tilde{R}$ such that

$$\tilde{R} = e^{-\Phi} \left(R - (d-1)G^{\mu\nu}\tilde{\nabla}_\mu\tilde{\nabla}_\nu\Phi - \frac{(d-1)(d-2)}{4}G^{\mu\nu}\tilde{\nabla}_\mu\Phi\tilde{\nabla}_\nu\Phi \right) \quad (\text{A7})$$

where, as in the main text, $\tilde{\nabla}$ denotes the connection compatible with $G_{\mu\nu}$. As discussed in detail in section (VII), not all metric variations are admissible. In order to correctly deduce the field equations, we should temporarily relax the assumption that $G_{\mu\nu}$ is of fixed determinant. The calculation should then be performed as described by (7.10), in which we freely vary S , before subtracting a term proportional to the trace of this variation. Upon obtaining the final expression for the field equations, we may reinstate the condition $\det G = -1$. With these considerations, it follows that, upon integrating by parts and assuming no contribution from boundary terms, the relevant action in four dimensions may be expressed as

$$S = \frac{1}{2\kappa} \int d^4x \sqrt{-G} e^\Phi \left(\tilde{R} + \frac{3}{2}G^{\mu\nu}\partial_\mu\Phi\partial_\nu\Phi \right) \quad (\text{A8})$$

Variations of this action with respect to $G^{\mu\nu}$ yield

$$\delta S = \frac{1}{2\kappa} \int d^4x \sqrt{-G} e^\Phi \left(-\frac{1}{2} \left(\tilde{R} + \frac{3}{2}G^{\alpha\beta}\partial_\alpha\Phi\partial_\beta\Phi \right) G_{\mu\nu} \delta G^{\mu\nu} + \delta G^{\mu\nu} \tilde{R}_{\mu\nu} + G^{\mu\nu} \delta \tilde{R}_{\mu\nu} + \frac{3}{2} \delta G^{\mu\nu} \partial_\mu\Phi\partial_\nu\Phi \right)$$

The behaviour of $\tilde{R}_{\mu\nu}$ under a change $\delta G^{\mu\nu}$ is a standard result [18]

$$\delta \tilde{R}_{\mu\nu} = \tilde{\nabla}_\rho \delta \tilde{\Gamma}_{\mu\nu}^\rho - \tilde{\nabla}_\mu \delta \tilde{\Gamma}_{\rho\nu}^\rho \quad (\text{A9})$$

We thus need only focus on the manipulation of the following term

$$\delta S \supset \frac{1}{2\kappa} \int d^4x \sqrt{-G} e^\Phi \tilde{\nabla}_\rho \left(G^{\mu\nu} \delta \tilde{\Gamma}_{\mu\nu}^\rho - G^{\rho\nu} \delta \tilde{\Gamma}_{\lambda\nu}^\lambda \right) := I_1 - I_2$$

The variation of the connection coefficients may be expressed as

$$\delta \tilde{\Gamma}_{\mu\nu}^\rho = \frac{1}{2} G^{\alpha\lambda} \left(\tilde{\nabla}_\nu \delta G_{\mu\alpha} + \tilde{\nabla}_\mu \delta G_{\nu\alpha} - \tilde{\nabla}_\alpha \delta G_{\mu\nu} \right) \quad (\text{A10})$$

We may then write the integral I_1 as follows (throughout, we do not retain total divergences)

$$\begin{aligned} I_1 &:= \frac{1}{2\kappa} \int d^4x \sqrt{-G} e^\Phi \tilde{\nabla}_\rho G^{\mu\nu} \delta \tilde{\Gamma}_{\mu\nu}^\rho \\ &= -\frac{1}{4\kappa} \int d^4x \sqrt{-G} e^\Phi (\partial_\rho \Phi) G^{\mu\nu} G^{\alpha\lambda} \left(\tilde{\nabla}_\nu \delta G_{\mu\alpha} + \tilde{\nabla}_\mu \delta G_{\nu\alpha} - \tilde{\nabla}_\alpha \delta G_{\mu\nu} \right) \\ &= -\frac{1}{4\kappa} \int d^4x \sqrt{-G} e^\Phi (\partial_\rho \Phi) \left(2 \tilde{\nabla}_\nu (G^{\mu\nu} G^{\alpha\lambda} \delta G_{\mu\alpha}) - \tilde{\nabla}_\alpha (G^{\mu\nu} G^{\alpha\lambda} \delta G_{\mu\nu}) \right) \\ &= \frac{1}{4\kappa} \int d^4x \sqrt{-G} e^\Phi \left[2 \left(\partial_\nu \Phi \partial_\rho \Phi + \tilde{\nabla}_\nu (\partial_\rho \Phi) \right) G^{\mu\nu} G^{\alpha\lambda} \delta G_{\mu\alpha} - \left(\partial_\alpha \Phi \partial_\rho \Phi + \tilde{\nabla}_\alpha (\partial_\rho \Phi) \right) G^{\mu\nu} G^{\alpha\lambda} \delta G_{\mu\nu} \right] \\ &= -\frac{1}{4\kappa} \int d^4x \sqrt{-G} e^\Phi \left[2 \left(\partial_\mu \Phi \partial_\nu \Phi + \tilde{\nabla}_\mu (\partial_\nu \Phi) \right) - G^{\alpha\beta} \left(\partial_\alpha \Phi \partial_\beta \Phi + \tilde{\nabla}_\alpha (\partial_\beta \Phi) \right) G_{\mu\nu} \right] \delta G^{\mu\nu} \end{aligned}$$

In passing to the final line, we have used that $\delta G_{\alpha\beta} = -G_{\alpha\mu} G_{\beta\nu} \delta G^{\mu\nu}$. A series of entirely analogous manipulations are carried out, yielding the following form of I_2

$$I_2 = -\frac{1}{4\kappa} \int d^4x \sqrt{-G} e^\Phi G^{\alpha\beta} \left(\partial_\alpha \Phi \partial_\beta \Phi + \tilde{\nabla}_\alpha (\partial_\beta \Phi) \right) G_{\mu\nu} \delta G^{\mu\nu} \quad (\text{A11})$$

Consequently, we find that the term in δS containing $\delta \tilde{R}_{\mu\nu}$ may be expressed as

$$-\frac{1}{2\kappa} \int d^4x \sqrt{-G} e^\Phi \left[\left(\partial_\mu \Phi \partial_\nu \Phi + \tilde{\nabla}_\mu (\partial_\nu \Phi) \right) - G^{\alpha\beta} \left(\partial_\alpha \Phi \partial_\beta \Phi + \tilde{\nabla}_\alpha (\partial_\beta \Phi) \right) G_{\mu\nu} \right] \delta G^{\mu\nu} \quad (\text{A12})$$

With this, the full (unconstrained) variation of the action with respect to $G^{\mu\nu}$ reads

$$\begin{aligned} \delta S = \frac{1}{2\kappa} \int d^4x \sqrt{-G} e^\Phi & \left[\tilde{R}_{\mu\nu} - \frac{1}{2} \left(\tilde{R} - \frac{1}{2} G^{\alpha\beta} \partial_\alpha \Phi \partial_\beta \Phi - 2 G^{\alpha\beta} \tilde{\nabla}_\alpha (\partial_\beta \Phi) \right) G_{\mu\nu} \right. \\ & \left. + \frac{1}{2} \partial_\mu \Phi \partial_\nu \Phi - \frac{1}{2} \left(\tilde{\nabla}_\mu (\partial_\nu \Phi) + \tilde{\nabla}_\nu (\partial_\mu \Phi) \right) \right] \delta G^{\mu\nu} \end{aligned} \quad (\text{A13})$$

Since we would like the final field equations to be symmetric in the lower indices, we have symmetrised the final term. The two steps that remain are application of (7.10), and elimination of $\partial_\mu \Phi$ in favour of the action density. For the former, we reproduce the relevant expression for the admissible variations

$$\frac{\delta S}{\delta^A G^{\mu\nu}} = \frac{\delta S}{\delta G^{\mu\nu}} - \frac{1}{4} G^{\mu\nu} G^{\alpha\beta} \frac{\delta S}{\delta G^{\alpha\beta}} \quad (\text{A14})$$

From this, we find that the equation of motion in conformal variables reads

$$\tilde{R}_{\mu\nu} - \frac{1}{4} \left(\tilde{R} + \frac{1}{2} G^{\alpha\beta} \partial_\alpha \Phi \partial_\beta \Phi - G^{\alpha\beta} \tilde{\nabla}_\alpha (\partial_\beta \Phi) \right) G_{\mu\nu} + \frac{1}{2} \partial_\mu \Phi \partial_\nu \Phi - \frac{1}{2} \left(\tilde{\nabla}_\mu (\partial_\nu \Phi) + \tilde{\nabla}_\nu (\partial_\mu \Phi) \right) = 0$$

Finally, the velocities $\partial_\mu \Phi$ of the conformal factor are replaced with the appropriate action-dependent expressions, leading to

$$\begin{aligned} \tilde{R}_{\mu\nu} - \frac{1}{4} \left(\tilde{R} + \frac{2\kappa^2}{9} G_{\alpha\beta} s^\alpha s^\beta - \frac{2\kappa}{3} \partial_\alpha s^\alpha \right) G_{\mu\nu} + \frac{2\kappa^2}{9} G_{\mu\alpha} G_{\nu\beta} s^\alpha s^\beta \\ - \frac{\kappa}{3} \left(G_{\mu\lambda} \tilde{\nabla}_\nu s^\lambda + G_{\lambda\nu} \tilde{\nabla}_\mu s^\lambda \right) = 0 \end{aligned} \quad (\text{A15})$$

precisely as in (7.12).

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